

Review Questions

- Start a new Colab file: open colab.research.google.com.
 - Write text answers in text blocks.
 - Write code answers in code blocks and run them.
 - Submit the URL to your file in Canvas ("Python review I").
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1. What are the main steps of a typical machine learning workflow?
2. Write Python code to create two lists: one for hours studied and one for scores. Then print both lists. Sample output:
[1, 2, 3, 4, 5]
[50, 55, 65, 70, 80]
3. Write Python code to compute the average (mean) of the scores list **without using external libraries**. Tip: to sum a list, use `sum()`, to get the length of a list, use `length()`. Sample output: 64.0
4. Ask Gemini (in a text cell): "What relationships might be interesting in a dataset with variables 'hours studied' and 'exam score'?"
Then write Python code to create a simple scatter plot of the data and run it. Add a `title()` to the plot.
5. Modify the previous code to compute and print the predicted score using a simple linear rule: $\text{prediction} = 40 + 8 * \text{hours}$.
Print `prediction` as a function of `hours` using a `for` loop.
Ask Gemini for the exact syntax of a `for` loop in Python.
Sample output:
Hours: 1 Predicted score: 48
Hours: 2 Predicted score: 56
Hours: 3 Predicted score: 64
Hours: 4 Predicted score: 72
Hours: 5 Predicted score: 80
6. Ask Gemini: "What is overfitting and underfitting in simple terms?"
Then answer in one sentence in your own words (no code). Do not copy Gemini's answer verbatim.